

Techniques from Number Theory

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Problem Solving Seminar

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Warm up

Challenge

Determine the last digit of 7^{2026} .

Strategy:

- start with small powers of 7
- look for a pattern

Warm up

n	last digit of 7^n
1	7
2	9
3	3
4	1
5	7
6	9
7	3
8	1
$\vdots$	$\vdots$

Warm up

Conjecture:

The last digit seems to behave periodically with period 4.

$$7^{2026} = 7^{4 \cdot 506 + 2}$$

Therefore the last digit should be 9.

How do we **prove** it?.

Outline

- Modular math
- Euler's Theorem
- Binomial Theorem
- Chinese Remainder Theorem
- Putnam-style problems

Modular arithmetic

Definition:

Let a, b, n be integers.

We say $a \equiv b \pmod n$ if n divides $b - a$.

Examples:

- $2 \equiv 0 \pmod 1$
- $5 \equiv 2 \pmod 3$
- $-2 \equiv 5 \pmod 7$

Modular arithmetic

Properties:

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

- $a + c \equiv b + d \pmod{n}$
- $a - c \equiv b - d \pmod{n}$
- $ac \equiv bd \pmod{n}$
- $a^m \equiv b^m \pmod{n}$ for any integer $m \geq 0$

Examples

Example:

Since $8 \equiv 1 \pmod{7}$,

$$8^{99} \equiv 1^{99} = 1 \pmod{7}.$$

Example:

Since $20 \equiv -1 \pmod{7}$,

$$(3 + 20)^{10} \equiv 2^{10} \pmod{7}.$$

Also,

$$2^{10} = 2^{3 \cdot 3 + 1} = (2^3)^3 2 = 8^3 2 \equiv 1^3 2 = 2 \pmod{7}.$$

Back to warmup

Challenge

Determine the last digit of 7^{2026} .

Solution

The last digit is the value modulo 10.

$$7^2 = 49 \equiv -1 \pmod{10}.$$

Hence $7^4 \equiv 1 \pmod{10}$, and

$$7^{2026} = (7^4)^{506} 7^2 \equiv 1^{506} \cdot 49 \equiv 9 \pmod{10}.$$

Therefore the last digit is 9!

Digging deeper

Question:

Why did the last digit of 7^n repeat every four times?

Answer:

It was because $7^4 \equiv 1 \pmod{10}$. Therefore

$$7^{n+4} = 7^n 7^4 \equiv 7^n \pmod{10}.$$

Question:

Yeah, but why is $7^4 \equiv 1 \pmod{10}$? Coincidence?

Outline

- Modular math
- **Euler's Theorem**
- Binomial Theorem
- Chinese Remainder Theorem
- Putnam-style problems

Euler's totient function

Euler's totient function $\varphi(n)$ counts the integers from 1 to n that are relatively prime to n . It satisfies

- $\varphi(1) = 1$
- $\varphi(p^n) = (p - 1)p^{n-1}$
- $\varphi(ab) = \varphi(a)\varphi(b)$ if $\gcd(a, b) = 1$

Examples:

- $\varphi(5) = 5 - 1 = 4$
- $\varphi(10) = \varphi(2 \cdot 5) = \varphi(2)\varphi(5) = 4$
- $\varphi(25) = \varphi(5^2) = (5 - 1)5^1 = 20$

Euler and Fermat

Euler's Theorem:

If $n > 0$ and $\gcd(a, n) = 1$ then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Special case:

Fermat's Little Theorem:

If p is prime and doesn't divide a , then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Back to warmup – again

Challenge

Determine the last digit of 7^{2026} .

Solution

By Euler's Theorem

$$7^{\varphi(10)} = 7^4 \equiv 1 \pmod{10}.$$

Consequently

$$7^{2026} = 7^{4 \cdot 506 + 2} = (7^4)^{506} 7^2 \equiv 1^{506} 7^2 = 49 \pmod{10}.$$

Therefore the last digit is 9!

First Problem

Challenge

Determine the last three digits of 3^{2026} .

First Problem

Solution start

We want the value of 3^{2026} modulo 1000.

We calculate

$$\varphi(1000) = \varphi(2^3 5^3) = \varphi(2^3) \varphi(5^3) = 4 \cdot 100 = 400.$$

Therefore by Euler's Theorem

$$3^{2026} = 3^{400 \cdot 5 + 26} = (3^{400})^5 3^{26} \equiv 3^{26} \pmod{1000}.$$

We still have to deal with 3^{26} ...

First Problem

Solution finish

Squaring strategy

$$3^2 \equiv 9 \pmod{1000}$$

$$3^4 = 9^2 \equiv 81 \pmod{1000}$$

$$3^8 = 81^2 \equiv 561 \pmod{1000}$$

$$3^{16} = 561^2 \equiv 721 \pmod{1000}$$

$$3^{26} = 3^{16} 3^8 3^2 \equiv 721 \cdot 561 \cdot 9 \equiv 329 \pmod{1000}.$$

The last three digits are 329!

Outline

- Modular math
- Euler's Theorem
- **Binomial Theorem**
- Chinese Remainder Theorem
- Putnam-style problems

Binomials

For integers $0 \leq k \leq j$, we define the **binomial**

$$\binom{j}{k} = \frac{j!}{k!(j-k)!} = \frac{j(j-1)(j-2)\dots(j-k+1)}{k(k-1)(k-2)\dots 1}.$$

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Binomials

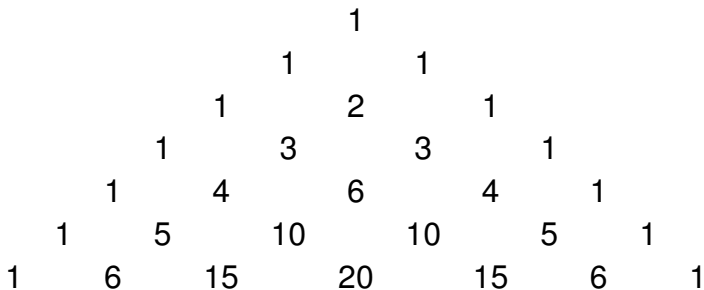
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- $\binom{j}{0} = 1$ and $\binom{j}{j} = 1$ for all $j \geq 0$
- $\binom{j}{1} = j$ for all $j \geq 1$
- $\binom{j}{2} = \frac{j(j-1)}{2}$ for all $j \geq 1$

Pascal's Triangle



Each entry is the sum of the two entries directly above it:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Binomial Theorem

Binomial coefficients allow us to expand powers of sums.

Binomial Theorem

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

Example:

$$(2 + 5)^3 = \binom{3}{0} 2^0 5^3 + \binom{3}{1} 2^1 5^2 + \binom{3}{2} 2^2 5^1 + \binom{3}{3} 2^3 5^0.$$

$$(2 + 5)^3 = 1 \cdot 2^0 5^3 + 3 \cdot 2^1 5^2 + 3 \cdot 2^2 5^1 + 1 \cdot 2^3 5^0.$$

$$(2 + 5)^3 = 125 + 150 + 60 + 8 = 343.$$

First Problem – another solution

Challenge

Determine the last three digits of 3^{2026} .

Solution

$$3^{2026} = 9^{1013} = (10 - 1)^{1013} = \sum_{k=0}^{1013} \binom{1013}{k} 10^k (-1)^{1013-k}$$

Modulo 1000 only the first three terms survive!

$$3^{2026} \equiv -\binom{1013}{0} + \binom{1013}{1} 10 - \binom{1013}{2} 10^2 \pmod{1000}$$

$$3^{2026} \equiv -1 + 10130 - 1013 \cdot 506 \cdot 100 \pmod{1000}$$

$$3^{2026} \equiv -1 + 130 - 800 = -671 \equiv 329 \pmod{1000}$$

- Modular math
- Euler's Theorem
- Binomial Theorem
- **Chinese Remainder Theorem**
- Putnam-style problems

A remainder puzzle

Challenge

Find an integer x satisfying

$$x \equiv 2 \pmod{3} \quad \text{and} \quad x \equiv 3 \pmod{5}.$$

Strategy:

- numbers congruent to 2 modulo 3 are

$$2, 5, 8, 11, 14, \dots$$

- the first one congruent to 3 modulo 5 is 8

Thus $x \equiv 8 \pmod{15}$.

Chinese Remainder Theorem

Theorem

Suppose $\gcd(m, n) = 1$. For any integers a and b , the system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

has exactly one solution modulo mn .

In other words, knowing the remainder modulo m and the remainder modulo n determines the remainder modulo mn .

How to construct the solution

To solve

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n},$$

start with the first congruence and write

$$x = a + mk.$$

Substitute this into the second congruence:

$$a + mk \equiv b \pmod{n}.$$

Since $\gcd(m, n) = 1$, the number m has an inverse modulo n , so we can solve for k .

An example

Challenge

Solve

$$x \equiv 4 \pmod{7}, \quad x \equiv 5 \pmod{9}.$$

Write $x = 4 + 7k$. Then

$$4 + 7k \equiv 5 \pmod{9},$$

so $7k \equiv 1 \pmod{9}$. Since $7^{-1} \equiv 4 \pmod{9}$,

$$k \equiv 4 \pmod{9}.$$

Therefore

$$x \equiv 4 + 7 \cdot 4 = 32 \pmod{63}.$$

Splitting a large modulus

Recall our problem of finding 3^{2026} modulo 1000. Since

$$1000 = 8 \cdot 125 \quad \text{and} \quad \gcd(8, 125) = 1,$$

the Chinese Remainder Theorem says it is enough to calculate separately modulo 8 and modulo 125.

Modulo 8

Since $3^2 \equiv 1 \pmod{8}$,

$$3^{2026} \equiv 1 \pmod{8}.$$

The calculation modulo 125

Since $\varphi(125) = 100$, Euler's Theorem gives

$$3^{2026} \equiv 3^{26} \pmod{125}.$$

Repeated squaring gives

$$3^8 \equiv 61 \pmod{125}, \quad 3^{16} \equiv 96 \pmod{125}.$$

Consequently

$$3^{26} = 3^{16} 3^8 3^2 \equiv 96 \cdot 61 \cdot 9 \equiv 79 \pmod{125}.$$

Putting the remainders together

We now know

$$x \equiv 1 \pmod{8}, \quad x \equiv 79 \pmod{125}.$$

Write $x = 79 + 125k$. Reducing modulo 8 gives

$$7 + 5k \equiv 1 \pmod{8}.$$

Thus $5k \equiv 2 \pmod{8}$, so $k \equiv 2 \pmod{8}$. Therefore

$$x \equiv 79 + 125 \cdot 2 = 329 \pmod{1000}.$$

The last three digits are again 329.

When the moduli are not coprime

The coprime assumption matters.

General rule

The system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

has a solution if and only if

$$a \equiv b \pmod{\gcd(m, n)}.$$

For example, $x \equiv 1 \pmod{4}$ and $x \equiv 3 \pmod{6}$ have the solution $x \equiv 9 \pmod{12}$, while $x \equiv 0 \pmod{4}$ and $x \equiv 1 \pmod{6}$ have no solution.

Practice

Solve each system.

① $x \equiv 3 \pmod{5}$ and $x \equiv 4 \pmod{7}$

② $x \equiv 1 \pmod{4}$ and $x \equiv 4 \pmod{9}$

③ $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, and $x \equiv 2 \pmod{7}$

Extension

Use the Chinese Remainder Theorem to find the last two digits of 7^{222} by working modulo 4 and modulo 25.

Universal fifth powers

Challenge

Prove that for every integer n ,

$$30 \mid n^5 - n.$$

Useful tools

Fermat's Little Theorem and the Chinese Remainder Theorem.

Universal fifth powers

Solution

We check the prime factors 2, 3, and 5 separately.

- Modulo 2, the integers n and n^5 have the same parity.
- Modulo 3, either $3 \mid n$, or $n^2 \equiv 1 \pmod{3}$, so $n^5 \equiv n \pmod{3}$.
- Fermat's Little Theorem gives $n^5 \equiv n \pmod{5}$.

Thus 2, 3, and 5 all divide $n^5 - n$. Since they are pairwise coprime, the Chinese Remainder Theorem gives

$$30 \mid n^5 - n.$$

A larger universal divisor

Challenge

Prove that for every integer n ,

$$2730 \mid n^{13} - n.$$

Hint

Factor 2730 and compare 12 with $p - 1$ for each prime divisor p .

A larger universal divisor

Solution

We have

$$2730 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 13.$$

For each prime p in this list, $p - 1$ divides 12. If $p \nmid n$, Fermat's Little Theorem gives

$$n^{12} \equiv 1 \pmod{p},$$

and hence $n^{13} \equiv n \pmod{p}$. The same conclusion is immediate when $p \mid n$.

Therefore every prime factor of 2730 divides $n^{13} - n$. Since the prime factors are distinct,

$$2730 \mid n^{13} - n.$$

Square roots of one

Challenge

How many incongruent solutions does

$$x^2 \equiv 1 \pmod{1001}$$

have?

Extension

Find all of the solutions modulo 1001.

Square roots of one

Solution

Factor the modulus:

$$1001 = 7 \cdot 11 \cdot 13.$$

For each prime $p \in \{7, 11, 13\}$,

$$x^2 \equiv 1 \pmod{p}$$

has exactly two solutions: $x \equiv 1 \pmod{p}$ and $x \equiv -1 \pmod{p}$.

We may choose either sign independently for each of the three primes. The Chinese Remainder Theorem turns each choice into exactly one residue modulo 1001. Hence the number of solutions is

$$2 \cdot 2 \cdot 2 = \boxed{8}.$$

A power tower

Solution

The exponent is odd, so $7^{7^7} \equiv 7 \pmod{8}$. Since $\varphi(125) = 100$ and $7^7 \equiv 43 \pmod{100}$,

$$7^{7^7} \equiv 7^{43} \equiv 93 \pmod{125}.$$

Write $x = 93 + 125k$. The congruence $x \equiv 7 \pmod{8}$ becomes

$$5 + 5k \equiv 7 \pmod{8},$$

which gives $k \equiv 2 \pmod{8}$. Therefore

$$x \equiv 93 + 125 \cdot 2 = \boxed{343} \pmod{1000}.$$

A binomial congruence

Challenge

Let p be prime. Prove that for every positive integer r ,

$$(1 + x)^{p^r} \equiv 1 + x^{p^r} \pmod{p}.$$

What does this tell us about

$$\binom{p^r}{k} \quad \text{for } 0 < k < p^r?$$

A binomial congruence

Solution

For $0 < k < p$, the prime p divides $\binom{p}{k}$, so the binomial theorem gives $(1 + x)^p \equiv 1 + x^p \pmod{p}$. Repeating this congruence,

$$\begin{aligned}(1 + x)^{p^r} &= \left((1 + x)^{p^{r-1}} \right)^p \\ &\equiv \left(1 + x^{p^{r-1}} \right)^p \\ &\equiv 1 + x^{p^r} \pmod{p}.\end{aligned}$$

Comparing coefficients shows that $p \mid \binom{p^r}{k}$ whenever $0 < k < p^r$.

Putnam 2000, B5

Begin with a finite set S_0 of positive integers. For $n \geq 0$, define S_{n+1} by the rule

$$a \in S_{n+1}$$

if and only if exactly one of $a - 1$ and a belongs to S_n .

Challenge

Prove that there are infinitely many positive integers N for which

$$S_N = S_0 \cup \{N + a : a \in S_0\}.$$

Hint

Encode the sets by polynomials whose coefficients are reduced modulo 2.

Putnam 2000, B5

Polynomial encoding

Work with coefficients modulo 2 and define

$$F_n(x) = \sum_{a \in S_n} x^a.$$

The coefficient of x^a in $(1+x)F_n(x)$ counts, modulo 2, how many of a and $a-1$ belong to S_n . It equals 1 precisely when exactly one of them belongs to S_n . Thus

$$F_{n+1}(x) = (1+x)F_n(x),$$

and consequently

$$F_N(x) = (1+x)^N F_0(x).$$

Putnam 2000, B5

Choosing N

Take $N = 2^r$. The binomial congruence gives

$$(1+x)^N \equiv 1 + x^N \pmod{2}.$$

Therefore

$$F_N(x) = F_0(x) + x^N F_0(x).$$

If $N > \max S_0$, the two groups of exponents are disjoint. The first polynomial records S_0 , while the second records

$$\{N + a : a \in S_0\}.$$

Every sufficiently large power of 2 therefore has the required property, and there are infinitely many such N .

Putnam 2018, B3

Challenge

Find all positive integers $n < 10^{100}$ satisfying all three conditions

$$n \mid 2^n,$$

$$n - 1 \mid 2^n - 1,$$

$$n - 2 \mid 2^n - 2.$$

Useful fact

For positive integers a and b ,

$$2^a - 1 \mid 2^b - 1 \iff a \mid b.$$

Putnam 2018, B3

First reduction

Since $n \mid 2^n$, the integer n has no odd prime factor. Hence $n = 2^m$. The last condition excludes $n = 2$, so $m \geq 2$. The second condition becomes

$$2^m - 1 \mid 2^{2^m} - 1.$$

By the useful fact, this is equivalent to $m \mid 2^m$. Therefore m is itself a power of 2. Writing $m = 2^\ell$ with $\ell \geq 1$, we obtain

$$n = 2^{2^\ell}.$$

Putnam 2018, B3

Second reduction

Divide the third condition by 2:

$$2^{m-1} - 1 \mid 2^{n-1} - 1.$$

Hence $m - 1 \mid n - 1$. Substituting $m = 2^\ell$ and $n = 2^{2^\ell}$ gives

$$2^\ell - 1 \mid 2^{2^\ell} - 1.$$

Applying the useful fact again, we obtain

$$\ell \mid 2^\ell.$$

Thus ℓ must also be a power of 2.

Putnam 2018, B3

The size bound

Since

$$n = 2^{2^\ell} < 10^{100},$$

we have $2^\ell < 100 \log_2(10) < 333$. The powers of 2 satisfying this bound are

$$\ell = 1, 2, 4, 8.$$

Therefore the solutions are

$$4, \quad 16, \quad 65536, \quad 2^{256}.$$